Winston Li

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Builds systems from scratch across PL, ML, and the web.

Education

UC Irvine • B.S. in Computer Science w/ Honors, 3.9 / 4.0 GPA — Sep 2024 – Jun 2026

- graduated in 2 years • ICS Honors Program • UROP Fellow
- Data Structures • Algorithms • Software Eng. • Compilers • Cryptography
- Formal Lang. & Automata • Quantum Computation • Distributed Systems

Projects

manifold • *Rust, wgpu, CEF, Servo* — Sep 2026 – Present

- Built an 80k-line browser: an engine process per tab (Blink, Servo, WebKit) over IPC, with hot-swap.
- Added crash recovery, journaled sessions, and a force-directed tab graph on **wgpu**.

scratchtape • *Rust, wgpu, cubecl* — Sep 2026 – Present

- Built an ML engine from scratch (autodiff, layers, GPU backend); every primitive is gradient-checked.
- Measured DX12 at 1.6–5.8× lower overhead than Vulkan on small matmuls; up to 900 GFLOP/s.

tatic • *Rust, WebAssembly* — Mar 2026 – Present

- Built a **JIT** caching **WebAssembly** by content hash; fuzz-tested it and fixed a benchmark-found leak.

cereal • *C99* — Sep 2026 – Present

- Wrote a C99 front end and language server; matches gcc-13 on 3,908 of 3,910 runnable gcc.dg tests.

Apps & web • *TypeScript, Svelte, Tauri* — Sep 2026 – Present

- **arrow-game**: strict-TS browser roguelite, no runtime dependencies; 400+ tests on Bun, Node. Evolved a bot that survives 3× longer on held-out seeds to balance it.
- **lollipoppy**: Tauri (Rust) + Svelte desktop app that drives the League client.

Skills and interests

Proficient in Rust, C / C++, JS / TS, Python, Svelte, Tailwind.

Familiar with Git, HTML / CSS, Nix, LaTeX, Typst, Claude Code, Unix systems & shells.

Acquainted with most mainstream languages, e.g., C#, Go, Haskell, Java, Lua, Ruby, SQL.

Domains: compilers & PL, ML & GPU compute, browsers, web & desktop apps, developer tooling.

Experience

Learning Assistant • UC Irvine — Mar 2025 – Jun 2026

- Delivered supplementary pedagogy for **Python** and upper-div **Algorithms,** e.g., Master Theorem.
- Spearheaded study sessions to identify and guide students through gaps in understanding.

Intern • LGBTQIA+ Identity Commission @ UC Irvine — Oct 2024 – Jun 2025

- Led a team of 3 to organize and host themed Gayme Show™.

Programmer • Video Game Development Club @ UC Irvine — Sep 2024 – May 2025

- Worked with team of 13 to create typing game in **C#** and **Unity**; led dev of the main UI window.